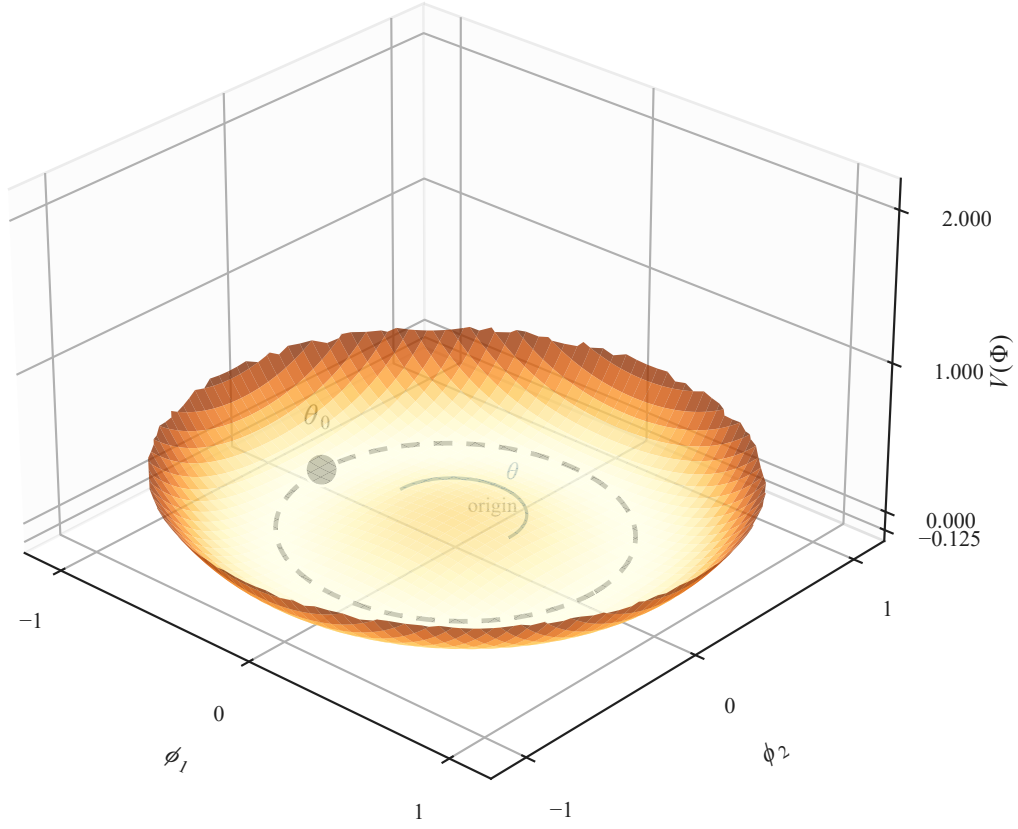
The Higgs Potential

Two Readings of the Same Surface

Standard QFT:
$V = -m^2|\Phi|^2 + \lambda|\Phi|^4$
(postulated)

GCT:
strain energy density of
breathing mode
(derived from lattice elasticity)



Phase Legend

Symmetric Phase
(at the origin): $T > T_c$

Broken Symmetry Phase
(ring of minima): $T < T_c$

Radial direction:
$|\Phi|$ = breathing mode
= physical Higgs boson

Angular direction:
θ = Goldstone bosons
→ longitudinal W and Z modes

Spontaneous Symmetry Breaking $SU(2) \rightarrow$ Gauge Coupling